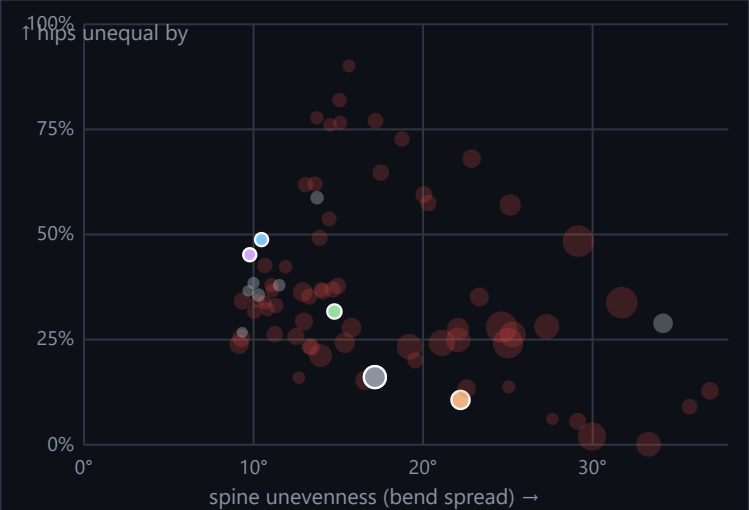


D-T2 · 5.1c — 80 samples, and all three goals pull apart

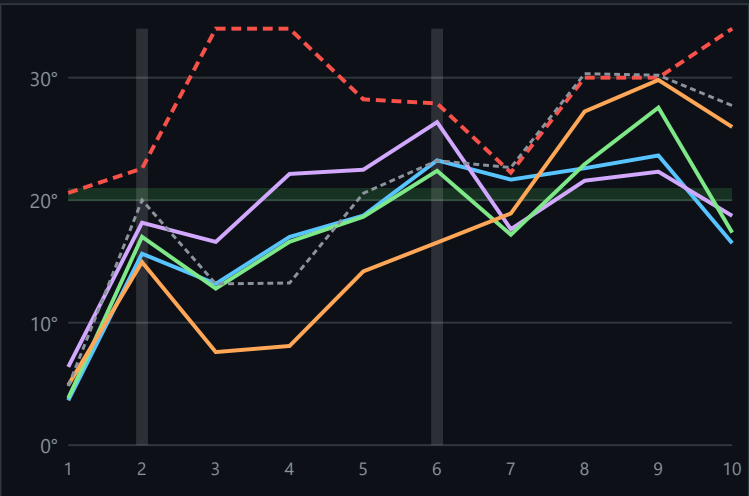
48 across the range, then 32 refined into the one basin that kept every joint legal. Fixed everywhere: MuJoCo, drag on, head excluded, legs 0.1 kg, no thrust yet.

Every sample: even spine vs equal hips



Each dot is one 12-second run; bigger dot = faster. Faint red = a joint hit its limit (56 of 80). **Bottom-left is the goal and it is empty.** Getting the spine even (moving left) makes the hips less equal (moving up) and slows the body down (dots shrink).

How much each joint bends, nose → tail



Red dashed = what the printed model physically allows. Green band = the 20–21° target. Pale bars mark the two hip joints. Grey dashed = where we were. Note segment 7's 22° limit sitting between neighbours allowed 28° and 30° — that dip is why 56 of 80 runs clipped there.

The four worth looking at

A · evenest spine

sections all at or below 1
push 0.39 · muscle 14 · sections 1/1/0.7/0.5/0.3
nothing at its cap
bend **13.2–23.6°**, spread **10.5°**, mean 19.1°
hips **15.6° → 23.3°** (ratio 1.49)
speed **0.32 u/s** swing 1.34 drift 0.1%
roll 9.3/s at peak 1.19°

B · evenest, shoulder pushed past 1

the last-resort option, for comparison
push 0.39 · muscle 16 · sections 1/1.3/0.5/0.35/0.35
nothing at its cap
bend **16.6–26.4°**, spread **9.8°**, mean 20.7°
hips **18.2° → 26.4°** (ratio 1.45)
speed **0.32 u/s** swing 1.25 drift 1.0%
roll 11.8/s at peak 1.39°

C · most headroom

furthest from every cap
push 0.39 · muscle 14 · sections 1/0.8/0.7/0.7/0.3
nothing at its cap
bend **12.8–27.6°**, spread **14.8°**, mean 19.2°
hips **17.0° → 22.4°** (ratio 1.32)
speed **0.40 u/s** swing 1.66 drift 0.6%
roll 10.6/s at peak 1.29°

D · most equal hips

hips within 11%
push 0.5 · muscle 14 · sections 0.8/1/1/0.5/0.5
nothing at its cap
bend **7.6–29.8°**, spread **22.2°**, mean 18.1°
hips **14.9° → 16.5°** (ratio 1.11)
speed **0.65 u/s** swing 1.60 drift 0.4%
roll 7.3/s at peak 0.39°

E · where we were

clips 3 joints — reference only
push 0.39 · muscle 18 · sections 1/1/1/0.6/0.6
CLIPS seg7, seg8, seg9
bend **13.2–30.3°**, spread **17.2°**, mean 22.3°
hips **20.0° → 23.2°** (ratio 1.16)

speed **0.88 u/s** swing 1.97 drift 0.5%
roll 6.8/s at peak 0.80°

Pushing a section above 1 bought almost nothing. B does it and lands within a degree of A, which keeps every section at or below 1 — same speed, slightly worse hips. The preference costs nothing.

Speed is the price. Nothing legal beats the old 0.89 u/s; the even shapes run 0.32–0.65. That was expected — thrust is the next stage's job, not the wave's.